

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		moles HCl = $0.090 \times 0.02265 = 2.04 \times 10^{-3}$ (1) moles NaOH unused in 25 cm ³ sample = 2.04×10^{-3} moles NaOH in volumetric flask = 2.04×10^{-2} (1) initial moles NaOH = $1.00 \times 0.050 = 5 \times 10^{-2}$ moles NaOH reacted = $5.00 \times 10^{-2} - 2.04 \times 10^{-2} = 2.96 \times 10^{-2}$ (1) allow ecf		2	1	3	2	3
		(ii)		moles (NH ₄) ₂ SO ₄ used = 1.48×10^{-2} mass (NH ₄) ₂ SO ₄ = $1.48 \times 10^{-2} \times 132.18 = 1.96$ g (1) percentage (NH ₄) ₂ SO ₄ = $\frac{1.96}{4.24} \times 100 = 46.2$ % (1) allow ecf from part (i) and for any calculated mass		2		2	1	
				Question 7 total	4	7	4	15	5	13